



The use of weight-of-evidence approaches to characterize developmental toxicity risk for therapeutic monoclonal antibodies in humans without *in vivo* studies

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ABSTRACT

Regulatory guidance for global drug development relies on animal studies to evaluate safety risks for humans, including risk of reproductive toxicity. Weight-of-evidence approaches (WoE) are increasingly becoming acceptable to evaluate risk. A WoE for developmental risk of monoclonal antibodies (mAbs) was evaluated for its ability to retrospectively characterize risk and to determine the need for further *in vivo* testing based on the remaining uncertainty. Reproductive toxicity studies of 65 mAbs were reviewed and compared to the WoE. Developmental toxicities were absent in 52/65 (80%) mAbs. Lack of toxicity was correctly predicted in 29/52 (56%) cases. False positive and equivocal predictions were made in 9/52 (17%) and 14/52 (27%) cases. For 3/65 (5%) mAbs, the findings were equivocal. Of mAbs with developmental toxicity findings (10/65, 15%), the WoE correctly anticipated pharmacology based reproductive toxicity without any false negative predictions in 9/10 (90%) cases, and in the remaining case (1/10, 10%) an *in vivo* study was recommended due to equivocal WoE outcome. Therefore, this WoE approach could characterize presence and absence of developmental risk without animal studies. The current WoE could have reduced the need for developmental toxicity studies by 42% without loss of important patient information in the label.

1. Introduction

The evaluation of developmental and reproductive toxicity (DART) risk for an investigational new drug relies on *in vivo* studies using pregnant animals. Global guidance to establish potential reproductive risk recommends embryofetal development (EFD) toxicity studies in rodents and rabbits, and pre- and postnatal development (PPND) toxicity studies in one rodent species, to understand the potential for small molecule drugs to cause malformations or embryofetal lethality (MEFL) or other adverse pregnancy outcomes (ICH S5(R3), 2020).

The intrinsic characteristics of biopharmaceuticals necessitate a different approach. For one, non-human primates (NHP) are often the only relevant model for toxicity testing. Furthermore, the placental transfer of monoclonal antibodies (mAbs) in the first trimester is low compared to the second and third trimester in both NHPs and humans

(Bowman et al., 2013; Pentsuk and van der Laan, 2009). The placental transfer of mAbs requires active transport via the binding of the Fc region of mAbs to the FcRn receptor expressed on placenta (Bowman et al., 2013; Moffat et al., 2014). Therefore, for this class of drug, stand-alone EFD studies with mAbs in NHP, which assess embryofetal toxicity during fetal organogenesis, are unlikely to provide useful human-relevant information, as toxicity may not manifest due to the very limited drug exposure during the first trimester. A separate guidance, ICH S6(R1), which takes into account the intrinsic properties of mAbs (ICH S6(R1), 2011), provides recommendations on the conduct of reproductive toxicity testing for this class of molecules. Because of the low placental transfer rates, an enhanced pre- and postnatal development (ePPND) study is recommended in NHP since the revision of the guideline ICH S6 in 2011 (ICH S6(R1), 2011) which adds EFD surrogate endpoints to a PPND study and is powered to allow for a more reliable evaluation of developmental risk (Jarvis et al., 2010). EFD endpoints are evaluated in

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Abbreviations

ADCs	antibody-drug conjugates
ATC	Anatomical Therapeutic Chemical
DART	Developmental and reproductive toxicity
EFD	Embryofetal development
EMA	European Medicines Agency
EPARs	European public assessment reports
ePPND	Enhanced pre- and postnatal development
INN	International Nonproprietary Name
MAA	Marketing Authorization Application
MEB	Medicines Evaluation Board;
MEFL	Malformations or embryofetal lethality
mAbs	Monoclonal antibodies
NHP	Non-human primates
PPND	Pre- and postnatal development
SmPC	Summary of product characteristics
WoE	Weight-of-evidence

newborns and infants instead of the fetus (Weinbauer et al., 2011). No additional studies are recommended beyond the ePPND to characterize developmental risk. Limitations of NHP as a test species for reproductive toxicity studies include low fertility and high spontaneous abortion rate (Chellman et al., 2009), single offspring, and the low statistical power to detecting infrequent adverse events (Jarvis et al., 2010). Because of the low placental transfer, dosing in the ePPND is initiated when pregnancy is confirmed, which generally corresponds to gestational day 18–20. This may theoretically limit the exposure during critical periods of embryo development occurring in implantation to a drug product.

However, whether *in vivo* data is necessary for every mAb to assess and communicate developmental toxicity risk has been argued (Bowman, 2016; Rocca et al., 2018). Due to their high target specificity, the safety profile of mAbs is generally favorable and predictable, with most of the toxicity consisting of exaggerated pharmacology or immune reactions. Toxicity of unknown etiology rarely occurs (Bugelski and Treacy, 2004; Chien et al., 2023; van Meer et al., 2013). Furthermore, phenotypic data from knockout mice have recently been shown to be highly predictive of MEFL in EFD studies on biopharmaceuticals (Catlin

et al., 2021). Therefore, one may question whether the developmental toxicity risk assessment of mAbs, which demonstrate exclusive target specificity and a relatively benign safety profile, must always be demonstrated in an animal study. In addition to scientific considerations, since the COVID-19 pandemic, availability and supply of mature NHP has been substantially reduced due to a prohibition on exporting NHP from China, limiting testing capacity (Mellet and Pepper, 2021). In addition to ethical concerns, these factors drive a strong need for alternative approaches to assess DART risk of mAbs.

The use of WoE assessment as an alternative approach to demonstrate the presence of DART risk of biopharmaceuticals is already noted as a regulatory acceptable option in global guidance such as ICH S6(R1). Alternative approaches including assessment of literature, placental transfer, and/or other factors might be considered appropriate for biopharmaceuticals intended for oncology indications if scientifically justified (ICH S9, 2009). For general biopharmaceuticals not limited to oncology indications, ICH S6(R1) allows using a WoE approach when the mechanism of action, phenotypic data from genetically modified animals, and class effects indicated presence of a DART risk (ICH S6(R1), 2011). Bowman and Chapin provided a similar framework for determining what new *in vivo* data are required for DART assessment by assessing the likelihood of off-target effects, risk tolerance, knowledge of target involvement in causing DART, and pharmacokinetics or exposure (Bowman and Chapin, 2016). Previously, Rocca et al. identified WoE characteristics, considering target biology, class effect, and pharmacokinetics, to evaluate reproductive risk of biopharmaceuticals (Rocca et al., 2018).

In this study, we used the method developed by Rocca et al. (Section 2.2.3) to evaluate whether a WoE approach can be applied to mAbs to identify for which mAbs an ePPND study should be conducted to assess developmental risk, and to determine in which cases the WoE would be sufficient to demonstrate presence or absence of risk.

2. Materials and methods

2.1. Data collection

2.1.1. Data selection

Monoclonal antibodies (mAbs) with a Marketing Authorization Application (MAA) were identified from the European Medicines

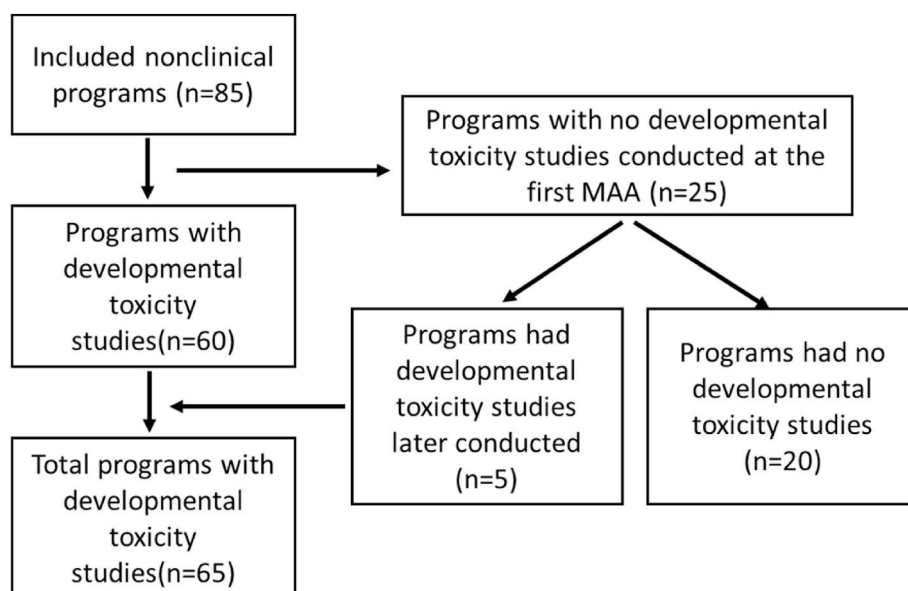


Fig. 1. Data inclusion/exclusion.
MAA: marketing authorization application.

	WoE assessment of DART risk										Face-valid arguments for waiving DART studies					
	Literature data on target biology					Class effect		Exposure		WoE						
	MEFL / G	Impaired organ function	FM tolerance	Uterine physiology	Placental toxicity	Foreign target	Non-clinical	Clinical	Systemic	Embryofetal	Conclusion	No <i>in vivo</i> model available	Teratogenic co-med	Not WOCBP	Advanced cancer	Short-term use
mAb 1																
mAb 2																
mAb 3																
mAb 4																
mAb 5																
mAb 6																
mAb 7																
mAb 8																
mAb 9																
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Fig. 2. (No) Weight-of-evidence assessment conducted by the applicant for cases where no *in vivo* studies were conducted, and the arguments based on risk-benefit considerations that the regulatory authorities accepted for omitting DART studies.

Red = presence of risk; yellow = inconclusive assessment; green = absence of risk; dark grey = argument due to benefit-risk considerations for regulatory acceptance other than WoE; White/empty = risk factor/argument not identified; Co-med = co-medication.

Agency (EMA) website (<https://www.ema.europa.eu/en/medicines/download-medicine-data>) irrespective of their regulatory status (authorized, refused, or withdrawn). All antibodies (excluding biosimilars) intended for human use, submitted for MAA up to August 1st, 2021, were collected. Biosimilars were excluded as they are copies of innovator molecules and the EU do not require animal studies if qualitative similarity is demonstrated and never require reproductive and developmental toxicity testing. Diagnostics and antibody-drug conjugates (ADCs) were further excluded from the dataset since these products are not intended to have a pharmacological effect and that ADCs have different risk profile from mAbs because most of the toxicity is generated by the cytotoxic chemical to which it is linked. The non-clinical program of the remaining therapeutic mAbs focusing on their developmental

toxicity component was included and analyzed.

2.1.2. Dataset

The dataset consisted of three main sections: 1) basic product information, 2) developmental toxicity study information, 3) regulatory implications and decisions on the developmental toxicity programs. The basic information of the included products, which was extracted from the European public assessment reports (EPARs), included the trade name; International Nonproprietary Name (INN); indication; Anatomical Therapeutic Chemical (ATC) code; current status; marketing authorization date; date of opinion; target; and modality. Product specific data on developmental toxicity studies was retrieved from Medicines Evaluation Board (MEB) review documents and included study

Table 1

Program designs.

Number of Studies per program	# Study type(s)		Species used	# mAbs
Non-rodent only (n = 43 mAbs)				
1	1 EFD	–	C	10
	1 EFD	–	Rb	2
	–	1 ePPND	C	19
2	–	2 ePPND	C	1
	1 EFD	1 PPND	C	5
	1 EFD	1 ePPND	C	4
	1 EFD (Rb)	1 ePPND	Rb + C	1
3	2 EFD	1 PPND	C	1
Rodent + non-rodent (n = 10 mAbs)				
2	1 EFD (C)	1 PPND	M ^a + C	1
	1 EFD (R)	1 ePPND	R + C	1
	1 FEED-EFD (M ^a)	1 ePPND	M ^a + C	1
3	2 EFD (M, Rb)	1 PPND (M)	M + Rb	1
	2 EFD (Mt)	1 PPND (M)	M + Mt	1
	2 EFD (Rb, R)	1 FEED-PPND (R)	R + Rb	1
	1 EFD (Rb) + 1FEED-EFD (R)	1 PPND (R)	R + Rb	1
4	1 EFD (Rb) + 2 FEED-EFD (R)	1 PPND (R)	R + Rb	1
	2 EFD (M ^a , C)	2 PPND (M ^a , C)	M ^a + C	1
	3 EFD (C, G)	1 ePPND	G + C	1
	Rodent only (n = 12 mAbs)			
1	1 EFD	–	M ^a	4
	1 EFD	–	M	1
	1 EFD	–	TG	1
	1 EFD	–	G	1
2	1 EFD	1 PPND	R ^a	1
	1 EFD	1 PPND	M ^a	3
3	1 EFD + 1 FEED-EFD	1 PPND	R	1

FEED-EFD = combined FEED and EFD study; FEED-PPND = combined FEED and PPND study; C = Cynomolgus monkey; Rb = Rabbit; M = Mouse (wild-type); R = Rat; Mt = Marmoset; TG = transgenic mouse; G = Guinea pig. In this dataset, for an EFD study the dosing typically started on gestation day (GD) 6 and ended on GD12 in rodent and between GD20 and GD50 in cynomolgus monkey, while the dosing started from 2 to 6 weeks prenatally to GD13 – GD20 for a combined FEED-EFD study in rodent. For a PPND study the dosing period typically ranged from GD6 to post-parturition day (PPD) 21 in rodent and from GD50 to parturition in cynomolgus. For an ePPND study the dosing period was from GD20 to parturition in cynomolgus monkey. For a combined FEED-PPND study the dosing started from 2 weeks prenatally to GD21 or PPD21.

^a Animal surrogate tested.

type; study year; species used; number of animal groups and group sizes; dosing period; and maternal, embryofetal, and/or postnatal toxicities when detected. If programs without DART studies were identified, the justification for not conducting a developmental toxicity study provided by the applicant was noted. In the final section, regulatory implications were recorded. This included regulatory assessments of the DART studies or regulatory opinions on the justification provided by the applicant for not conducting a developmental toxicity study.

2.2. Analysis

2.2.1. Programs with developmental toxicity studies

Regarding the programs of marketed mAbs containing developmental toxicity studies, maternal, embryofetal, and/or postnatal adverse outcomes were identified. When developmental toxicity was identified, the nature of the toxicity (i.e. pharmacology-related, potentially caused by maternal toxicities, and human relevance) was determined.

2.2.2. Programs containing no developmental toxicity studies

In order to investigate whether a WoE approach was used to

Table 2

Human relevance of *in vivo* developmental toxicity findings in DART studies.

Clinical relevance	#mAbs	(%)
Not of human concern	52	(80)
No noteworthy findings	4	
Not treatment related	27	
Treatment related, non-adverse	18	
Treatment related, species specific	3	
Equivocal	3	(5)
MEFL	2	
Other toxicity	1	
Of human concern	10	(15)
MEFL of unknown etiology	1	
MEFL attributed to pharmacology	8	
Other toxicity attributed to pharmacology	1	

communicate DART risk for mAbs and the associated regulatory opinions on the WoE justifications, the analysis of this group focused on: 1) scrutinizing the arguments for regulatory acceptance; 2) identifying whether a WoE approach was used for risk communication by the applicants; 3) comparison of the outcomes of WoE assessments and regulatory opinions stated in the regulatory assessment reports and the summary of product characteristics (SmPC). Arguments for regulatory acceptance such as patient populations and indications (Fig. 2) were not considered a part of the WoE approach.

2.2.3. WoE approach validation using programs with DART studies

The WoE approach proposed by Rocca et al. was used to characterize developmental risk of mAbs in our database (Rocca et al., 2018). Briefly, all available data including data generated in basic research, human and/or animal genetics, data derived from other molecules targeting similar pathways and/or surrogate molecules was reviewed. This included an evaluation of likelihood of embryofetal exposure, target biology, class effects, and availability of pharmacologically relevant test systems to evaluate developmental toxicity risk.

True positive predictions were defined as mAbs where developmental toxicity was correctly identified by the WoE compared to the *in vivo* outcome. False positive predictions were defined as mAbs where developmental toxicity was incorrectly identified by the WoE compared to the *in vivo* outcome. True negative predictions were defined as mAbs where the absence of developmental toxicity was correctly identified by the WoE compared to the *in vivo* outcome. False negative predictions were defined as mAbs where the absence of developmental toxicity was incorrectly identified by the WoE compared to the *in vivo* outcome.

Accuracy was established as the sum of true positive predictions, true negative predictions, and true equivocal prediction divided by the total number of mAbs. True positive prediction rate was calculated as the number of mAbs with positive effects observed in the *in vivo* studies divided by the total number of positive predictions. True negative prediction rate was established as the number of mAbs with negative effects observed in the *in vivo* studies divided by the total number of negative predictions. Sensitivity was calculated as true positive predictions divided by the total number of mAbs with reproductive toxicity identified in the DART studies. Specificity was calculated as true negative predictions divided by the total number of mAbs without reproductive toxicity identified in the DART studies.

This exercise was carried out by three independent researchers of this study. The weight of evidence was established based on (literature) data available at the time of product development.

3. Results

3.1. Data inclusion/exclusion

In total, 103 mAb human medicinal products excluding biosimilars that were applied for MAA (either authorized, refused, or withdrawn) up

Textbox 1**Adverse findings of human concern**

MEFL accounted for the most adverse findings of concern during pregnancy (9/10). However, 88% of the MEFL (8/9) was attributable to the target pharmacology, with only one case where the findings could not be attributed to the pharmacology of the molecule (Table 2).

MEFL of unknown etiology.

A PPND study in cynomolgus monkeys with ipilimumab revealed an increase in abortion, stillbirth, premature delivery associated with lower birth weight, and infant mortality (ipilimumab, Table 3). These findings could be attributed to the primary pharmacology common for immune checkpoint inhibitors. However, unilateral renal agenesis in one female neonate, and imperforate urethra in one male neonate were observed. These malformations could not be conclusively identified as treatment-related (Yervoy SmPC5.3, 2023), but were considered of human concern as urogenital malformation rarely occurs in cynomolgus monkeys (Peterson et al., 1997). A direct link with the action of ipilimumab as an immune checkpoint inhibiting mAb could not be made.

Toxicity other than MEFL.

A lasting inhibition of the immune system was observed in neonates exposed to efalizumab (efalizumab, Table 3). While this is not an MEFL, it is an adverse developmental effect mediated by the target pharmacology.

Textbox 2**Adverse findings considered equivocal**

For crizanlizumab, increased embryofetal loss was attributed to anti-drug antibodies (ADA)-mediated toxicity (Table 3). However, it is well established that immune responses in animals are not translatable to humans (Bugelski and Treacy, 2004; crizanlizumab, Table 3).

Species differences contributed to equivocal findings for evinacumab, since the rabbit has a significantly lower baseline lipid level compared with human. Fetal loss and/or premature delivery and fetal malformations were attributed to hypolipidemia. Finally, a dose-dependent minimal to mild glomerulopathy was noted in neonates exposed to ocrelizumab, which was not correlated to the pharmacological action. Similar products had not shown this adverse effect, and the clinical relevance was unknown. Nevertheless, the finding was considered equivocal by the researchers of this study, including two senior non-clinical assessors (evinacumab, ocrelizumab, Table 3).

to August 1st, 2021, were collected. Of these 103 medicinal products, 8 antibody drug conjugates and 7 diagnostic antibodies were excluded. Of the remaining 88 products meeting acceptance criteria (Section 2.1.1), three additional products were excluded as these were duplicate compounds marketed under different brands. Non-clinical development programs supporting 85 therapeutic mAbs were included into the final dataset. Of these 85 programs, 25 did not conduct DART studies, while 60 did. For 5 out of the 25 programs without DART studies, subsequent development of the product led to DART studies being conducted after the first marketing authorization. This was due to an application for a non-oncology indication ($n = 5$). These 5 non-clinical programs were added to the 60 programs with DART studies for analysis (Fig. 1).

3.2. Overview of dataset

Around 38% of the included programs where the mAb was developed for anti-neoplastic therapy, and 34% for immunosuppressant therapy. Anti-neoplastic agents accounted for 72% (18/25) of the programs without DART studies, while almost half of the programs containing DART studies were developed for immunosuppressants (47%, 28/60). The remainder of the products were developed for a range of other indications (Supplementary Fig. 1).

For 8% of the included programs where the mAb had no Fc component. Among the programs where the mAb contained an Fc component ($n = 78$), IgG1 accounted for 62% of the IgG subclass, followed by IgG4 (15%) and IgG2 (12%), with 2 mAbs that were IgG2 and IgG4 hybrid antibodies (IgG2/4).

Developmental toxicity testing using non-rodents accounted for 43/65 (66%) programs. For 40/43 (93%) programs, cynomolgus monkey was the only species used (Table 1). For 29/43 (67%) of these programs, the studies were conducted in or before 2010, with 2 (5%) programs for which the study date was not identified and 12 (28%) programs for

which the studies were conducted in or after 2011 when the ICH S6(R1) came into operation. Both rodent and non-rodent were used in 10/65 (15%) programs. For 7/10 (70%) programs using rodent and non-rodent, the studies were conducted in or before 2010, with 3 (30%) programs for which the studies were conducted in or after 2011. In 12/65 (18%) programs, rodents only were tested for developmental toxicity. Of 9/12 (75%) programs using only rodents for developmental toxicity testing, the studies were conducted in or before 2010, with studies conducted in or after 2011 in 3 (25%) programs. For 8/12 (67%) programs, the developmental toxicity was assessed using surrogate mAbs, which are homologs of the clinical candidate mAb, but are directed against the orthologous antigen in an animal model, typically rodent. In 4/12 (33%) programs, the clinical candidates were tested in rodents (Table 1).

The majority of the programs contained only 1 study ($n = 38$, 58%), followed by programs containing 2 studies ($n = 18$, 28%), 3 studies ($n = 6$, 9%), and 4 studies ($n = 3$, 5%). When a program consisted of 1 study only ($n = 38$), this was often an ePPND ($n = 19$, 50%), an EFD in cynomolgus monkeys ($n = 10$, 26%), or an EFD in mice testing mouse surrogates ($n = 4$, 11%) (Table 1).

3.3. Programs with DART studies ($n = 65$)

3.3.1. Adverse findings in DART studies

No adverse developmental findings or developmental findings of human concern were identified in the DART studies in 80% of the programs (52/65 cases, Table 2). Adverse developmental findings were identified in 13 programs (20%) of which adverse developmental findings were considered of human concern in 10 programs (15%, Table 2, Textbox 1). In 3 programs (5%) the adverse developmental findings were considered equivocal (Table 2, Textbox 2).

Table 3
Adverse findings in DART studies.

mAb	ATC code	Target	Findings		Species	Study type	Clinical relevance	PD-related?	ADA formation in dams	Impact of ADA
			Dams	Offspring						
Crizanlizumab	B06	P-selectin		Increase in fetal loss (abortions or stillbirths) (Adakveo_SmPC5.3, 2023)	Cynomolgus monkey	ePPND	Equivocal	No	Yes	Toxicity
Evinacumab	C10	ANGPTL3		Soft tissues and skeletal malformations Premature neonatal death, fetal loss and/or premature delivery (Evkeeza_SmPC5.3, 2023)	Rabbit (not a suitable species)	EFD	Equivocal	Yes	Yes	Exposure
Ocrelizumab	L04	CD20		Lymphoid follicle formation in bone marrow Lymphoplasmacytic renal inflammation Glomerulopathy ^a Decreased testicular weight Five confirmed or suspected infections partially impacted by B-cell depletion Depleted B cell populations in newborns (Ocrevus_SmPC5.3, 2023)	Cynomolgus monkey	ePPND	Equivocal	No	Yes	Potential toxicity in the dams in the EFD study (Ocrevus_EPAR, 2018)
Efalizumab	L04	CD11a		Decreased T-cell dependent immunity visible at 11 weeks of age (Raptiva_SmPC5.3, 2009)	Mouse ^b	PPND	Of human concern	Yes	Yes	No
Bevacizumab	L01	VEGF	Dose-dependent decrease in body weight	Decreased fetal body weights Increased specific gross and skeletal fetal malformations Increased mean number of fetal resorptions (Avastin_SmPC5.3, 2023)	Rabbit	EFD	Of human concern	Yes	Yes	Exposure
Olaratumab	L01	PDGFR α		Increased incidence of skeletal variations Eye malformation: (partially) open eyes (Lartruvo_SmPC5.3, 2019)	Mouse ^b	EFD	Of human concern	Yes	N/I	N/I
Pertuzumab	L01	Growth factor receptor HER2		Dose dependent Increased embryofetal death Kidney hypoplasia, lung hypoplasia, ventricular septal defects, thin ventricular wall Oligohydramnios (Perjeta_SmPC5.3, 2021)	Cynomolgus monkey	EFD	Of human concern	Yes	N/I	N/I
Panitumumab	L01	EGFR		Fetal abortions and/or fetal deaths (Vectibix_SmPC5.3, 2022)	Cynomolgus monkey	EFD	Of human concern	Yes	Yes	No
Cetixumab	L01	EGFR		Dose-dependent increased abortion (	Cynomolgus monkey	EFD	Of human concern	Yes		

(continued on next page)

Table 3 (continued)

mAb	ATC code	Target	Findings		Species	Study type	Clinical relevance	PD-related?	ADA formation in dams	Impact of ADA
			Dams	Offspring						
Nivolumab	L01	PD-1		Erbixum_SmPC5.3, 2022 Dose-dependent increase in fetal losses Increased fetal mortality (Opdivo_SmPC5.3, 2023)	Cynomolgus monkey	ePPND	Of human concern	Yes	Yes	Exposure
Burosumab	M05	FGF23	Moderate mineralization of the placenta	Shortening of the gestation period and associated increased incidence of premature births Slightly increased embryo-fetal deaths (high dose) (Crysivita_SmPC5.3, 2023)	Cynomolgus monkey	ePPND	Of human concern	Yes	Yes	No
Natalizumab	L04	$\alpha 4$ integrins		Small reduction in pup survival	Guinea pig	EFD	Of human concern	Yes	Yes	No
				Increased abortion Mild anemia, reduced platelet counts, increased spleen weights and reduced liver and thymus weights in fetus Reduced platelet count in neonates (Tysabri_SmPC5.3, 2023)	Cynomolgus monkey	EFD, ePPND			Yes	Exposure
Ipilimumab	L01	CTLA-4		Increased abortion, stillbirth, premature delivery (with corresponding lower birth weight), infant mortality Urogenital malformations observed in one male and female infant ^a (Yervoy_SmPC5.3, 2023)	Cynomolgus monkey	PPND	Of human concern	No	Yes	No

TG mouse = transgenic mouse; SmPC = summary of product characteristics; N/I = not identified.

^a finding of unknown etiology.

^b surrogate used.

3.3.2. Concordance between rodent and non-rodent developmental findings

When the mAb was tested in both rodent and non-rodent ($n = 10$, [Table 1](#)), for 8 programs the findings were concordant between these species. For one program, additional findings were seen in rabbits. The observed soft tissues and skeletal malformations, premature neonatal death, fetal loss and/or premature delivery after exposure to evinacumab was considered rabbit specific as this species has low baseline lipid levels during pregnancy (Evinacumab, [Table 3](#)). For the other program, increased abortion and anemia were seen in monkeys but not in guinea pigs (natalizumab, [Table 3](#)).

3.4. Programs without DART studies in the first MAA ($n = 25$)

3.4.1. Face-valid arguments for waiving DART studies

In 7 programs intended for non-oncology indications (mAb 11–16, and mAb 18, [Fig. 2](#)), face-valid arguments such as regulatory or benefit-risk considerations supported the regulatory acceptance of the absence of developmental toxicity studies. For 3 programs (mAb 13, 14, and 15, [Fig. 2](#)) the target of the mAb was exogenous (i.e. bacterial toxin or viral

protein) and studies were not recommended, in line with ICH S6(R1); for 2 programs (mAb 11 and 12, [Fig. 2](#)) the main patient population excluded women of child-bearing potential; for 1 program (mAb 18, [Fig. 2](#)) the mAb was intended for short-term use only; for 1 program (mAb 16, [Fig. 2](#)) no pharmacologically relevant models were available to conduct the reproductive toxicity study (dark grey cells, [Fig. 2](#)). The remaining 18 programs were developed for advanced cancer indications (in line with ICH S9), and for 5 of these (mAb 6, 17, 20, 22, and 24, [Fig. 2](#)), the clinical co-medication had already been identified as a potential teratogen, further supporting the lack of ePPND studies (dark grey cells, [Fig. 2](#)).

3.4.2. The use of WoE approach for DART assessment and risk communication

In 20/25 programs a WoE approach was used by the company to evaluate the potential for reproductive toxicity (mAb 1 - 20, [Fig. 2](#)).

An increased risk was predicted for 10 programs, which was reflected in the SmPC in 9 cases. Involvement of target biology; mode of action in embryofetal development or pregnancy; known class effects; placental

Table 4
Summary of the validation of WoE approach.

Total mAbs (n = 65)	WoE Predicted Yes	WoE Predicted No	WoE Predicted Equivocal
<i>In vivo</i> Yes	9	0	1
<i>In vivo</i> No	9	29	14
Equivocal <i>in vivo</i> findings	0	2	1

transfer; and fetal expression of the target, were recurring justifications. In 1/10 program (mAb 5, Fig. 2) the WoE was not labelled in the SmPC, likely due to the limited literature data.

Arguments for a low risk were identified in 6 programs. Low systemic and embryofetal exposure (n = 2); the target being exogenous (n = 3); absence of relevant systemic toxicity in general toxicity studies (n = 1) were recurring arguments to support the negative WoE. Generally, there is less regulatory acceptance of a WoE suggesting a lower risk. For 2 programs the WoE suggested a lower risk, but developmental risk remained to be labelled in the SmPC (mAb 11, 12, Fig. 2).

In four programs where the mAbs were intended for short-term use (n = 1) or advanced cancer indications (n = 3), an inconclusive WoE was considered sufficient for waiving an ePPND study.

3.5. Validation of WoE approach

Of the 65 programs, the WoE approach, correctly predicted the outcomes of *in vivo* studies for 39 programs (60% accuracy, Table 4). It

was identified as equivocal for 16 programs (Table 4, Fig. 3), presence of risk for 18 programs (Figs. 4 and 5), and absence of risk for 31 programs (ocrelizumab, crizanlizumab, Fig. 5; Supplementary Fig. 2). Retrospectively, this was correct in 9 programs for presence of risk (bevacizumab, panitumumab, cetuximab, natalizumab, burosumab, pertuzumab, olaratumab, nivolumab, ipilimumab, Figs. 5), 29 programs for absence of risk (Supplementary Fig. 2) and equivocal in 1 program (evinacumab, Fig. 5), leading to a true positive prediction rate of 50% and a true negative prediction rate of 94% (Table 4).

Accuracy (60%) was established as the sum of true positive predictions (n = 9, Fig. 5), true negative predictions (n = 29, Supplementary Fig. 2), and true equivocal prediction (n = 1, Fig. 5) divided by the total number of mAbs (n = 65). True positive prediction rate (50%) was calculated as the number of mAbs with positive effects observed in the *in vivo* studies (n = 9) divided by the total number of positive predictions (n = 18, Figs. 4 and 5). True negative prediction rate (94%) was established as the number of mAbs with negative effects observed in the *in vivo* studies (n = 29) divided by the total number of negative predictions (n = 31, Fig. 5, Supplementary Fig. 2). Sensitivity (90%) was calculated as true positive predictions (n = 9, Fig. 5) divided by the total number of mAbs with reproductive toxicity identified in the DART studies (n = 10). Specificity (56%) was calculated as true negative predictions (n = 29, Supplementary Fig. 2) divided by the total number of mAbs without reproductive toxicity identified in the DART studies (n = 52).

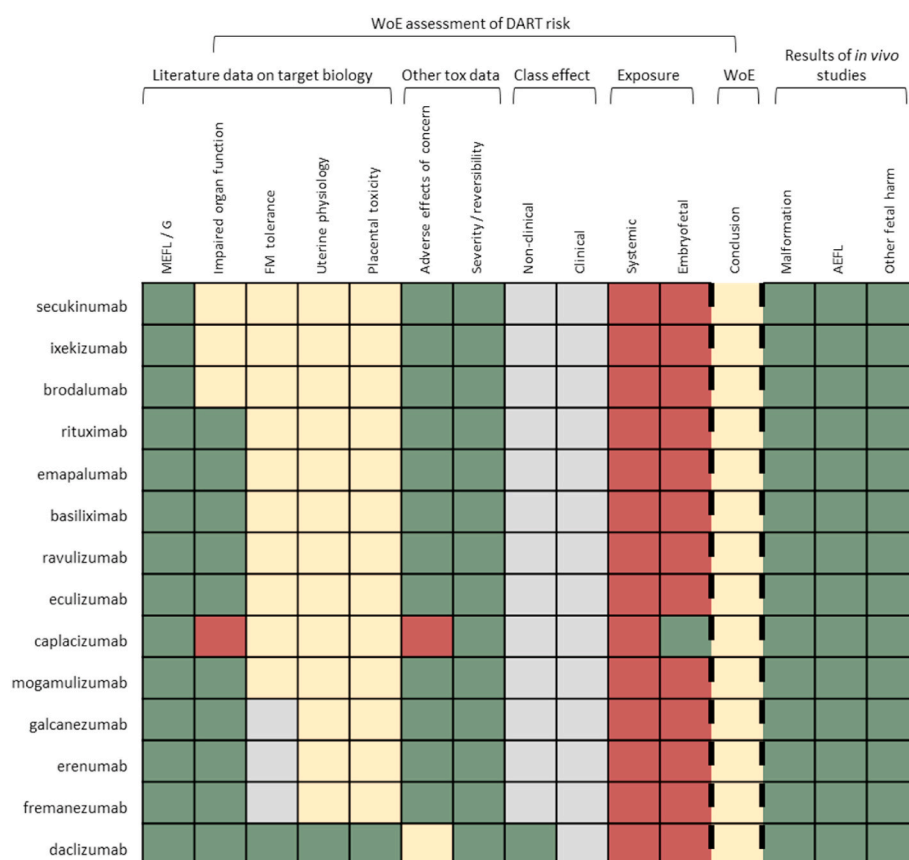


Fig. 3. Cases in which no developmental toxicities were identified in the *in vivo* studies were predicted as cases of equivocal risk based on the weight-of-evidence assessment.

The INN of mAbs and the weight-of-evidence assessment factors are displayed in the left and the top panel of Fig. 3, respectively. WoE = weight of evidence; DART = developmental and reproductive toxicity; MEFL/G = malformation and embryofetal lethality/growth retardation; FM tolerance = feto-maternal tolerance; M = malformation; AEFL = abortion or embryofetal lethality; tox = toxicity. Red = presence of risk; green = absence of risk; yellow = inconclusive assessment; light grey = no data or data not found.

4. Discussion

It is common knowledge that the vast majority of adverse effects observed in repeat-dose toxicity studies elicited by mAbs are caused by (exaggerated) pharmacology and/or immune responses (Baldrick, 2011; Bugelski and Treacy, 2004; Chapman et al., 2012; van Meer et al., 2013). Not surprisingly, these characteristic can be extended to reproductive toxicity findings with mAbs. DART of human concern was only noted in a small proportion of mAb programs (10/65, 15%). When DART of human concern was noted ($n = 10$), nearly all these adverse findings ($n = 9$) were anticipated based on the intended mechanism of action, and outcome of other general toxicity studies. Furthermore, most of the targets in which reproductive toxicity was noted were known to play a role in developmental biology or maintenance of pregnancy. In all other cases, reproductive toxicity did not occur when tested in pregnant animals, suggesting that the mode of action or target biology is a decisive factor for mAbs in predicting adverse pregnancy outcomes.

When intrinsic properties of a molecule drive a highly predictive safety profile, it raises the discussion if animal studies should always be conducted to confirm expected effects and to what extent unexpected and translatable findings would be identified. Over the last decade, scientific justifications to support the waiving of animal studies have increasingly been suggested or implemented for specific studies (Bowman and Chapin, 2016; Chien et al., 2023; ICHS 1B(R1), 2022; ICH S5(R3), 2020; ICH S11, 2020; Rocca et al., 2018). This further underlines that animal testing is not always required to evaluate and communicate risk and that for specific modalities or types of risk assessment, these approaches can be accepted by health agencies. Evidence based approaches can be employed for reproductive toxicity studies as well, which is outlined in ICH S6(R1) and ICH S5(R3). However, regulatory acceptance for waiving ePPND studies exists mostly for those molecules where risk can be demonstrated in a WoE exercise, which is generally reflected in the SmPC. Less known is the fact that a WoE based on absence of risk is a regulatory acceptable approach for

products intended for non-oncology indications. While current guidance does not recommend using a WoE based on negligible risk to justify obviating ePPND studies (ICH S6(R1), 2011), justifications to omit DART studies based on a negative WoE have been successfully submitted in different regions such as the European Union and the United States (Ackley et al., 2023) (Fig. 2). Therefore, on a case-by-case basis, there are scientifically justified arguments to omit reproductive toxicity studies in a non-clinical program based on both presence and absence of risk. When risk cannot be fully elucidated, studies in pregnant animals remain necessary. Not only is this in line with 3Rs principles, but it also reduces logistical difficulties of sourcing mature animals (Grimm, 2023). The regulatory acceptance of a WoE approach first requires the development of a robust model, which is then evaluated in a regulatory setting. Safe harbor principles may aid in generating experience and trust in such a model, for example by prospectively submitting a WoE assessment but to conduct the *in vivo* study in line with the established guidance, which was the approach used to develop the addendum for the revision of ICH S1B (Bourcier et al., 2024; ICH S1B(R1)). The results of these exercises could then lead to further discussions with stakeholders to evaluate the uptake of such a WoE approach in a revised guideline.

Rocca et al. have previously developed a WoE approach model to characterize developmental and reproductive toxicity and demonstrated its applicability using case studies. This model used the target biology and its role in development, maternal tolerance, uterine physiology, ovulation, and implantation as the critical factors to determine the developmental and reproductive risk of a molecule (Rocca et al., 2018). In our dataset, the mode of action as well as phenotypic profiles of genetically modified animals were risk factors that most often provided the scientific basis to characterize developmental risk.

The WoE approach was effective in separating mAbs with significant developmental risk (50% true positive prediction rate; 90% sensitivity) and those with evident absence of risk (94% true negative prediction rate; 56% specificity; 60% accuracy, Table 4) in our database.

WoE assessment of DART risk													
Literature data on target biology					Other tox data		Class effect		Exposure		WoE		Results of <i>in vivo</i> studies
MEFL / G	Impaired organ function	FM tolerance	Uterine physiology	Placental toxicity	Adverse effects of concern	Severity/reversibility	Non-clinical	Clinical	Systemic	Embryofetal	Conclusion	Malformation	AEFL
denosumab													
tildrakizumab													
guselkumab													
risankizumab													
canakinumab													
tralokinumab													
durvalumab													
trastuzumab													
romosozumab													

Fig. 4. Cases in which no developmental toxicities were identified in the *in vivo* studies were predicted as cases with presence of risk based on the weight-of-evidence assessment.

The INN of mAbs and the weight-of-evidence assessment factors are displayed in the left and the top panel of Fig. 4, respectively. WoE = weight of evidence; DART = developmental and reproductive toxicity; MEFL/G = malformation and embryofetal lethality/growth retardation; FM tolerance = feto-maternal tolerance; M = malformation; AEFL = abortion or embryofetal lethality; tox = toxicity. Red = presence of risk; green = absence of risk; yellow = inconclusive assessment; light grey = no data or data not found.

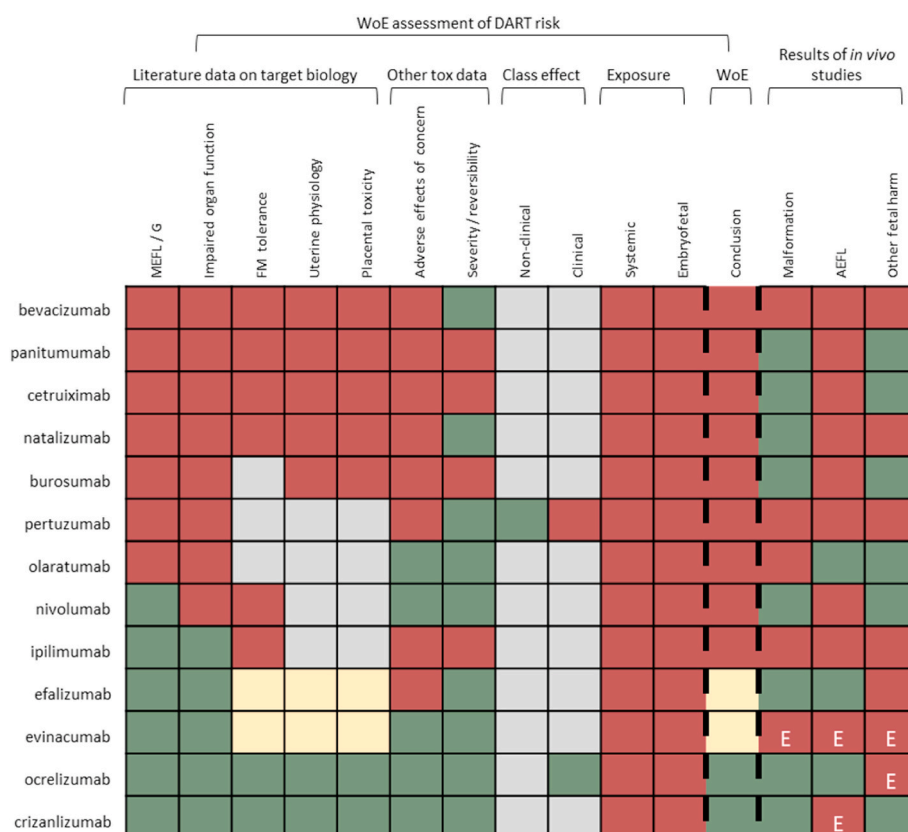


Fig. 5. Cases in which developmental toxicities or equivocal findings were identified in the *in vivo* studies were predicted as cases with presence of risk, absence of risk, or of equivocal risk, based on the weight-of-evidence assessment.

The INN of mAbs and the weight-of-evidence assessment factors are displayed in the left and the top panel of Fig. 5, respectively. WoE = weight of evidence; DART = developmental and reproductive toxicity; MEFL/G = malformation and embryofetal lethality/growth retardation; FM tolerance = fetomaternal tolerance; M = malformation; AEFL = abortion or embryofetal lethality; tox = toxicity. Red = presence of risk; green = absence of risk; yellow = inconclusive assessment; light grey = no data or data not found; E = equivocal findings were identified in the *in vivo* studies.

Importantly, in these cases the WoE exercise alone would have been sufficient to adequately communicate risk to patients in clinical trials or in the label. No false negative predictions were made using the WoE approach based on *in vivo* study outcome. This suggested that the use of the WoE approach would not put patients at risk while reducing animals studies. The relatively low true positive rate is likely a reflection of the limitations of this study, which was non-exhaustive. Overall, the WoE approach recommended additional animal studies in 16/65 (25%) of the programs to further evaluate DART risk. In these cases, there was either insufficient, missing or conflicting literature data, leading to an equivocal DART risk prediction. Nevertheless, even in those cases, the WoE model was too conservative in identifying products where an ePPND would be needed, with a false positive prediction rate of 50% (9/18 positive predictions by the WoE model). This can largely be explained by the fact that when literature suggested abortifacient effects after disruption of the mAb target, these effects were never observed in the NHP studies. Surprisingly, one of these cases was durvalumab, which targets the check-point protein PD-L1. Inhibition of PD-L1 is known to increase the rate of abortions as a result of decreasing fetomaternal tolerance, which was demonstrated in transgenic mouse studies (Gule-ria et al., 2005). The data from knock-out animal studies has been regularly used to justify the absence of developmental toxicity studies with other marketed anti-PD-(L)1 mAbs, and is reflected in the label. The conservative interpretation that resulted in more false positive cases also corresponded with an overreliance on literature, which was sometimes limited or ambiguous and further affected by limited resources of the authors of this article to comprehensively identify all relevant publications for each target or product. Therefore, based on the

literature evidence generated in this study, the WoE approach identified several other immunomodulatory mAbs as having high risk and not requiring further animal studies. However, this would potentially result in more conservative wording and risk minimization measures in the label, and limit treatment options for women of childbearing potential or pregnant women. In addition, our database contained products which were developed before ICH S6(R1), which used combinations of two-species testing and EFD-PPND-like designs, thus introducing difficulties in interpreting the study data.

Weight-of-evidence models which use more specific risk factors are likely to provide a more detailed understanding of the reproductive and developmental risks of a molecule and thus provide a better characterization of developmental and reproductive risk. For example, taking into account the biodistribution of reproductive tissues, or separating *in vitro* assays out as a specific factor to consider can aid in a more systematic and robust WoE assessment. More insight into abortifacient effects of immunomodulatory mAbs could provide a better WoE based safety profile of this class of antibodies. Nevertheless, and to conclude, the current model would have reduced the need for studies by 42% without once affecting patient safety or loss of important data in the label. Studies would be initiated in those cases where uncertainties remained or where risk could not be sufficiently translated to humans. We therefore propose that a WoE approach can provide scientifically-based arguments to demonstrate the presence or absence of developmental and/or reproductive risks that should be acceptable from a regulatory perspective and that are also in line with the 3Rs principles.

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CRedit authorship contribution statement

Hsiao-Tzu Chien: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Puck Roos:** Writing – original draft, Methodology, Formal analysis, Data curation. **Frans G.M. Russel:** Writing – review & editing, Supervision, Methodology. **Peter T. Theunissen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Peter J.K. van Meer:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.yrtph.2024.105682>.

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